

z5 (1.3)

Dane są liczby $a = 16^{\log_4 75}$ oraz $b = \frac{\log_{\sqrt{8}} 81}{\log_{64} 81}$.

Oblicz $a + 3b$.

① Obliczamy a : $a^{\log_a b} = b$ $r \cdot \log_a X = \log_a X^r$

$$a = 16^{\log_4 75} = 4^{2 \log_4 75} = 4^{\log_4 75^2} = 75^2 = 5625$$

② Obliczamy b :

$$b = \frac{\log_{\sqrt{8}} 81}{\log_{64} 81}$$

Mianownik: $\log_b c = \frac{\log_a c}{\log_a b}$

$$\log_{64} 81 = \frac{\log_8 81}{\log_8 64} = \frac{\log_8 81}{2}$$

Licznik:

$$\log_{\sqrt{8}} 81 = \frac{\log_8 81}{\log_8 \sqrt{8}} = \frac{\log_8 81}{\frac{1}{2}} = 2 \log_8 81$$

Cały ułamek:

$$b = \frac{2 \log_8 81}{\frac{\log_8 81}{2}} = 2 \log_8 81 \cdot \frac{2}{\log_8 81} = 2 \cdot 2 = 4$$

③ Obliczenie $a + 3b$:

$$a + 3b = 5625 + 3 \cdot 4 = 5625 + 12 = 5637$$

Odp: $a + 3b = 5637$.